

A road has to *kill* before anyone flags it.

Assam lost **3,296 people** on its roads in 2023. A stretch becomes an official blackspot only after **5 serious crashes or 10 deaths**. The warning arrives after the funerals.

OUR QUESTION

Which 500 m of road ahead is dangerous, and when does it get *worse*?

Answered from records that already exist. Tested on months the model never saw.

A car's sensors see about **200 m**, and only the present. Assam's crash record knows the next **2 km**, locked in **44,928 police PDFs**.



THE USER

Driver-assist & self-driving cars

They read the map ahead (curves, speed limits) but get nothing about danger. We add **one attribute: a risk band per 500 m**.

THE USER

Road-safety engineers (PWD, NHAI)

They must choose which stretches to fix **before they become blackspots**, not after five serious crashes.

THE EVIDENCE

Crashes are predictable

9% of all crashes sit on 570 spots of 100 m. Dark roads with no lights: **35% fatal** vs 23% by day.

Self-driving became possible once lanes were captured as data. We capture what lanes can't show: where, and when, roads hurt people.

Eight sources, one road record. *Every field tested before trusted.*

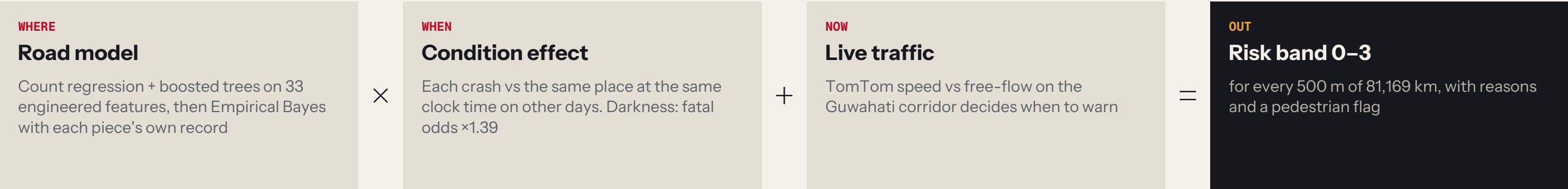


WHAT WE COLLECTED		KIND	USE
iRAD police reports	44,114 crashes · Jan 2023 – May 2026	OBS	LABELS
OpenStreetMap	246,789 road pieces of 500 m max	OBS	MODEL
NASA night lights	1.45 M pixels · street-light proxy	OBS	MODEL
ERA5 weather	3.79 M cell-hours, crash + control only	INF	MODEL
Sun position	193,904 place-hours, computed	COMP	MODEL
Airport METAR	50,618 Guwahati reports	OBS	CHECK
TomTom live traffic	19 tiles every 15 min + 24 points / h	OBS	LIVE ●
CE323 field studies	23 tables: counts, headways, speeds	OBS	FUTURE

Scoped to where and when crashes happened: **674** weather requests, not every hour of every cell. Every download is logged; everything is Parquet with a schema and samples. Sources we could not yet trust were **kept for checks and future work**, not thrown away.

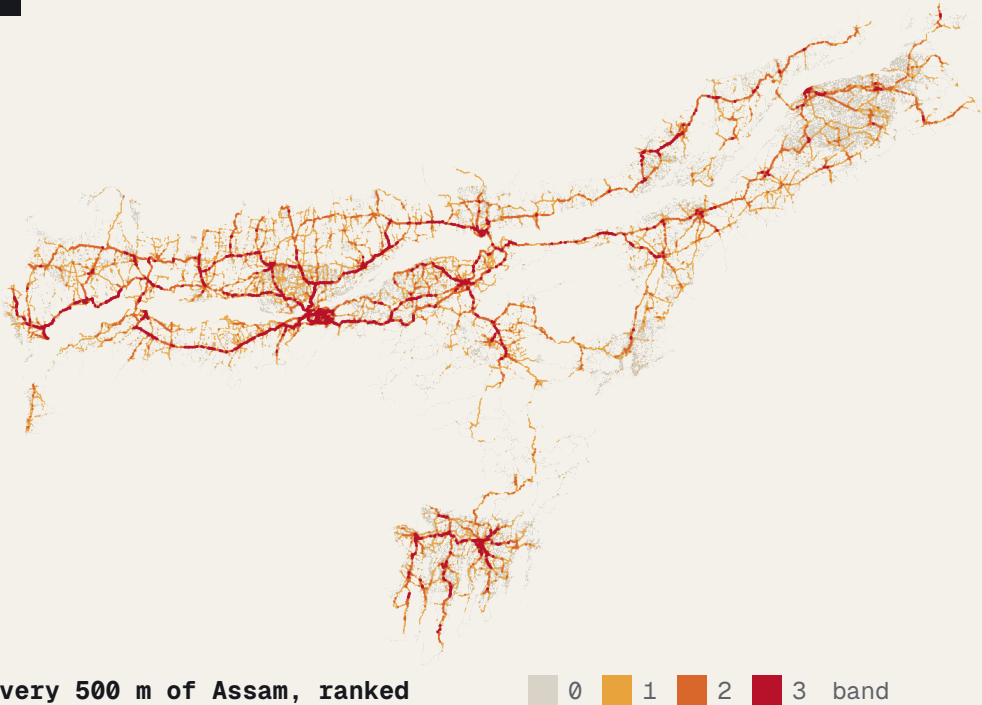
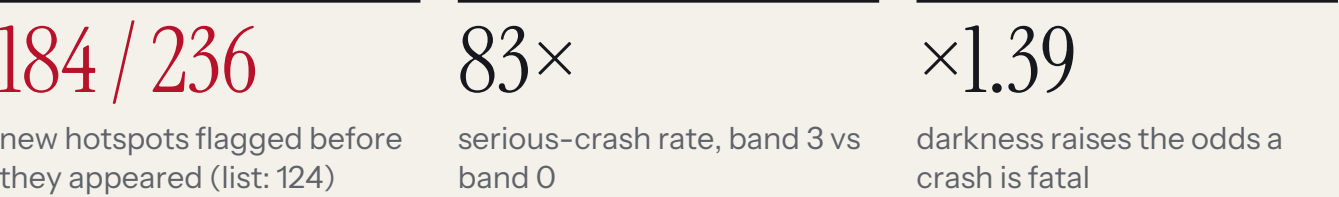
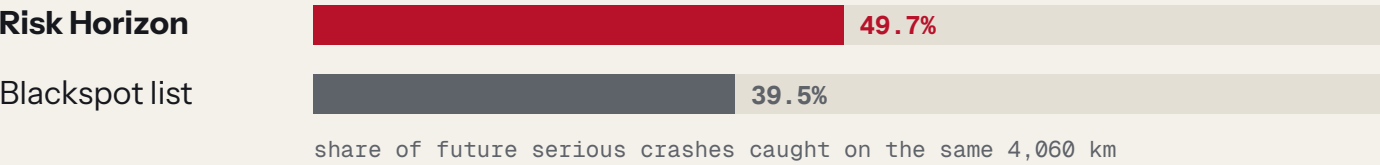
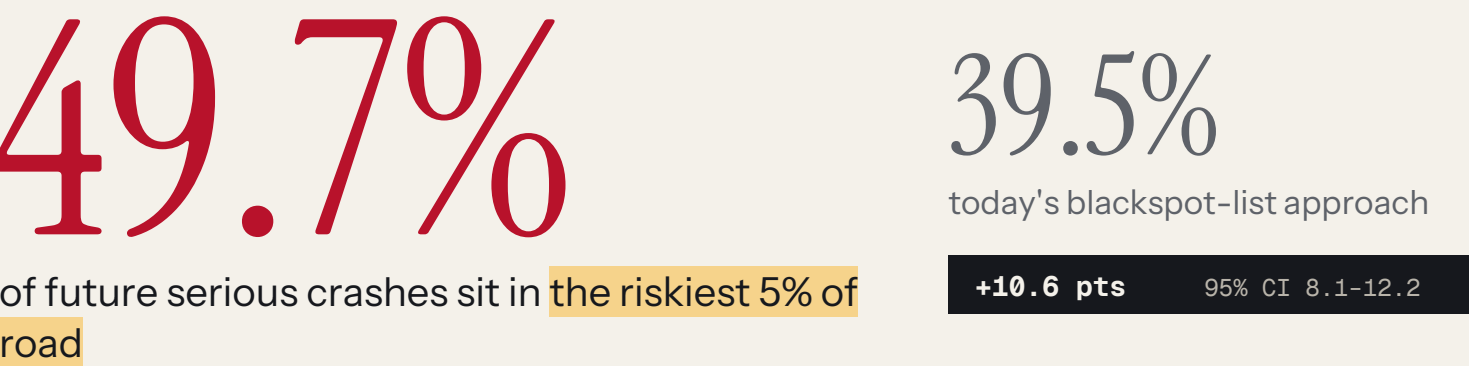
WHEN COLLECTION WENT WRONG

- iRAD use grew 1.5×, unevenly (0.8×–3.8×)
→ **drift-corrected features**
- Police "fog" matched sensors 11% vs 5%
→ **dropped; measured fog used**
- Behaviour fields under half filled
→ **never used as labels**
- No crashes logged Jan & Jul 2024
→ **rates per real data month**
- Crash points off the road
→ **snapped: 90% within 100 m**
- 2 hidden leaks in our own code
→ **caught by exact replication**



Half of future serious crashes, found on 5% of the road.

Trained Jan 2023 – Jun 2025 · judged once on Jul 2025 – May 2026, months it never saw



WHAT CHANGES

- BAND 3 · top 1%

Spoken warning 500 m ahead. Cruise slows before the stretch.
- BAND 2 · next 4%

Chime 300 m ahead. Speed capped, longer gap.
- AFTER DARK

Same roads move up a band.
- ENGINEERS

A ranked fix-list of 4,060 km, before the next blackspot.

WHAT OUR DATA CANNOT TELL YOU

No traffic volumes exist for Assam, so risk is per km, not per vehicle. Rain looks "safer" because fewer people travel, so a condition never lowers a warning. Single-crash severity is hard to predict (AUC 0.57).

NEXT

Live speeds + field counts → exposure. Post-crash alerts: 58% of deaths come after the first hour. Any state with iRAD + OSM.

● COLLECTION RUNNING NOW

TomTom every 15 min since 26 Sep

DEMO

<https://bit.ly/risk-horizon>